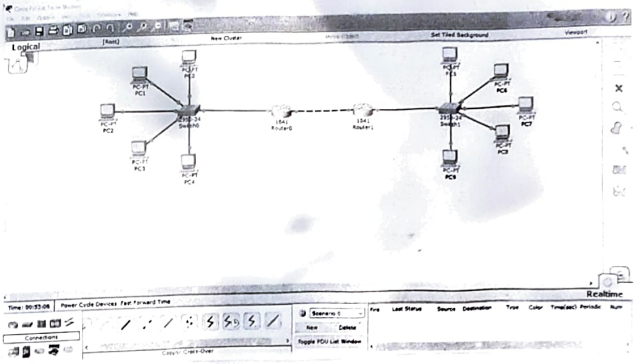
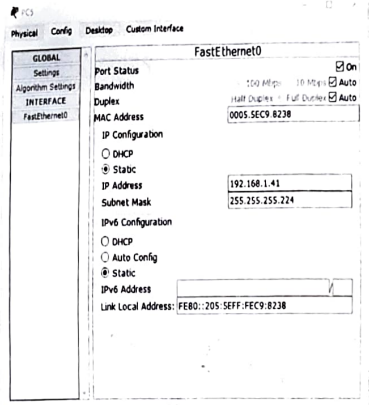
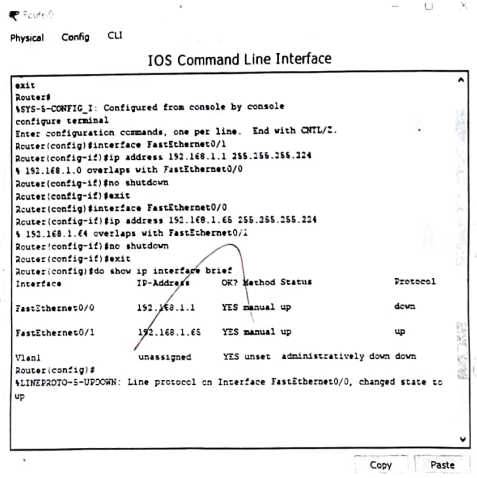






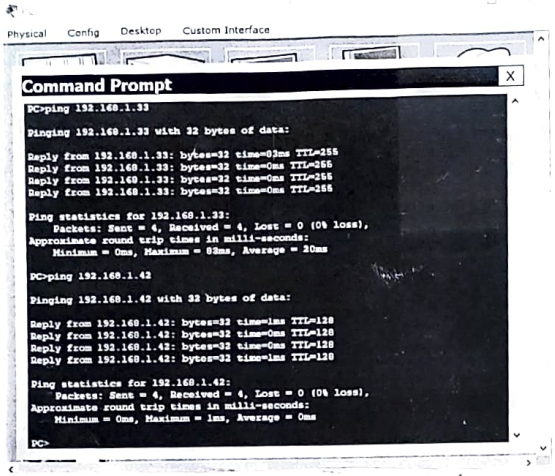
27/08/2024 EX:09 IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR

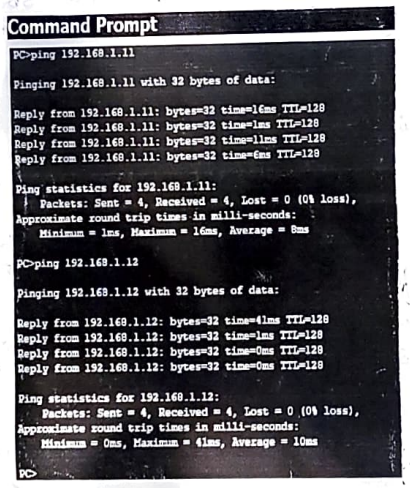
AIM:

To implement classless IP subnetting using Cisco Packet Tracer and Configure the network devices with appropriate IP addresses and subnet masks to establish communication between different subnets.

PROCEDURE:

1. In Cisco Packet Tracer create a new network topology.
2. Add the required network devices:
   - ⇒ 2 routers (R_0 and R_1)
   - ⇒ 2 switches (S_0 and S_1)
   - ⇒ 10 PC's (PC_0 - PC_9)
3. Connect PC_0 - PC_4 to switch S_0 and PC_5 - PC_9 to switch S_1 using suitable Ethernet cables.
4. Connect S_0 to R_0 and S_1 to R_1.
5. Connect R_0 and R_1 using their available fast-Ethernet interfaces.
6. Divide the network into /27 subnets. The subnet mask used is: 255.255.255.224.
7. Configure R_0 using the CLI. Assign 192.168.1.1 to the interface connected to S_0 and 192.168.1.65 to the interface connected to R_2
8. Enable the router interfaces using the no shutdown command.
9. Configure R_1 using CLI. Assign 192.168.1.33 to the interface connected to S_1 and 192.168.1.66 to the interface connected to R_0.
10. Enable the R_1 ifaces using the no shutdown command in cmd.
11. Configure the IP addresses of PC_0 - PC_4 in the 1st subnet
    (a) PC_0 - 192.168.1.17
    (b) PC_1 - 192.168.1.12





(c) PC_2 - 192.168.1.13
(d) PC_3 - 192.168.1.14
(e) PC_4 - 192.168.1.15

use subnet mask 255.255.255.224 and default gateway 192.168.1.1

12. Configure the Ip addresses of PC_5 - PC_9 in the 2nd subnet.

(a) PC_5 - 192.168.1.41
(b) PC_6 - 192.168.1.42
(c) PC_7 - 192.168.1.43
(d) PC_8 - 192.168.1.44
(e) PC_9 - 192.168.1.45

use subnet mask 255.255.255.224 and default gateway 192.168.1.33

13. Configure static routes on R_0 and R_1 so that packets can travel b/w the 2 LAN subnets.

14. Verify the router interfaces using the command: "show ip interface brief".

15. Test connectivity b/w PCs using the ping command from the command prompt.

16. Ping the default gateway from each PC and then ping PCs belonging to the other subnet.

17. If successful replies are received with no packet loss, the subnetted network is functioning correctly.

RESULT:

Thus, classless IP subnetting was successfully implemented using Cisco Packet Tracer, and connectivity between the subnetted networks was verified using the Ping command.